Training Day 22

1. Topics Covered and Brief Explanation

1. Revision of Artificial Intelligence: Revised the fundamentals of AI, its characteristics, importance, and applications in different industries.
2. Revision of Machine Learning: Reviewed supervised, unsupervised, and reinforcement learning along with regression, classification, and clustering.
3. Data Processing and Model Evaluation: Revised data preprocessing, dataset splitting, model training, prediction, and evaluation techniques such as accuracy, precision, recall, and F1-score.
4. Deep Learning: Reviewed artificial neural networks, their layers, neurons, weights, and activation functions.
5. Computer Vision: Revised image representation, image processing, object detection, and the use of OpenCV and YOLO.
6. Real-World Applications of AI and ML:
    - Healthcare – disease detection and medical image analysis.
    - Education – personalized learning systems.
    - Banking – fraud detection and risk analysis.
    - Transportation – traffic monitoring and autonomous vehicles.
    - Security – face recognition and surveillance systems.
7. Future Scope of AI and ML: Discussed how AI and ML are contributing to automation, smart systems, robotics, and intelligent applications.

2. Tools and Technologies Used

Tool/Technology	Purpose
Python	AI and ML programming
NumPy	Numerical computations
Pandas	Data manipulation
Matplotlib	Data visualization
Scikit-learn	Machine Learning algorithms
OpenCV	Computer Vision
YOLO	Object detection

3. Practical Learning / Code Examples

A simple example was used to revise the basic Machine Learning workflow, including dataset preparation, model training, and prediction.

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score

# Load dataset

data = load_iris()

X = data.data

y = data.target

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Train model

model = DecisionTreeClassifier()

model.fit(X_train, y_train)

# Make predictions

predictions = model.predict(X_test)

# Evaluate model

accuracy = accuracy_score(y_test, predictions)

print("Model Accuracy:", accuracy)
```

Explanation:

- The Iris dataset was loaded for classification.
- train_test_split() divided the data into training and testing sets.
- DecisionTreeClassifier() created a classification model.
- fit() trained the model using training data.
- predict() generated predictions for test data.
- accuracy_score() measured the model's performance.

4. Learning Outcomes

- Revised the fundamental concepts of AI, ML, and Deep Learning.

- Understood the basic Machine Learning development workflow.
- Gained familiarity with commonly used AI and ML tools.
- Explored computer vision and object detection techniques.
- Understood the practical applications of AI and ML across different industries.
- Developed a basic understanding of how intelligent systems are designed and implemented.